

School of Computer Science and Information Technology
Programme: BCA

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Network Administration [23BCA4VC02]

Activity – 2

[A Case Study Report on Phishing Attack Simulation]

Submitted by

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CERTIFICATE

This is to certify that **Mithan H S** has satisfactorily completed activity prescribed by JAIN (Deemed to be University) for the **Fourth Semester Degree Course** in the year 2025-2026.

Sl. No.	CRITERIA	MARKS	MARKS OBTAINED
1.	On-time Submission	10	
2.	Learning Outcomes	15	
3.	Report with course details and assessment	15	
4.	Viva	10	
	Total	50	
	Convert	15	

MARKS	
MAX	OBTAINED
15	

Signature of the Student:

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Case study: Phishing Attack Simulation

1 Introduction

Cybersecurity threats have become increasingly sophisticated, and one of the most common and dangerous threats is phishing. Phishing attacks attempt to trick users into revealing sensitive information by imitating trusted entities like websites or emails. This project simulates a phishing attack using Cisco Packet Tracer in a virtual network, without causing harm in the real world.

2. Background

Phishing is a type of cyberattack where attackers disguise themselves as legitimate entities to deceive users. Usually delivered via emails, phishing links lead users to fake login pages designed to steal credentials. This project uses Cisco Packet Tracer to demonstrate how such an attack can be simulated in a controlled lab environment for learning purposes.

Key concepts covered:

- Email spoofing
- Fake website creation
- Network simulation
- User credential harvesting (simulation only)

3. Discussion

This simulation shows how attackers can:

1. Send fake emails to victims.
2. Create realistic-looking login pages.
3. Harvest credentials when the victim interacts with the fake site.

By setting up email and web servers, and configuring a simple local area network, the attacker successfully tricks a victim into entering login information on a fake website. The demonstration proves how social engineering combined with technical setup can simulate real-world phishing.

4. Challenges:

While creating the simulation, a few challenges were encountered:

- Ensuring correct IP addressing and domain names across the network.
- Making the fake web page appear realistic.
- Configuring email services and SMTP/POP3 settings.
- Linking form actions in HTML to simulate credential capture.
- Correct usage of file names (`.html`, `hacked.txt`) to avoid file not found errors.

Troubleshooting these issues helped in understanding the internal working of web and mail protocols.

5. Implementation:

Devices Used:

- PC0 (Victim)
- PC1 (Attacker)
- Server0 (Mail Server)
- Server1 (Fake Web Server)
- Switch (2960)
- Router (optional)

IP Configuration:

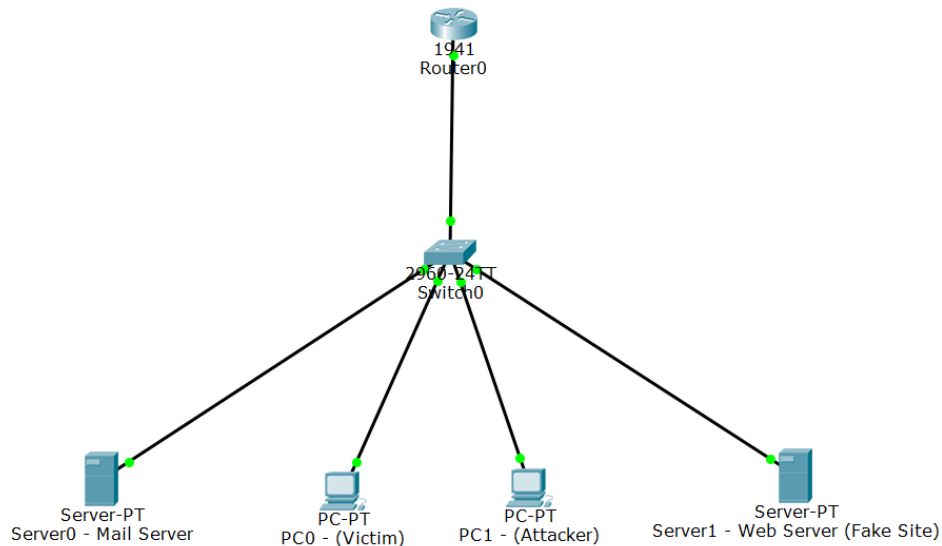
Device	IP Address	Role
PC0	192.168.1.10	Victim
PC1	192.168.1.20	Attacker
Server0	192.168.1.30	Mail Server
Server1	192.168.1.40	Fake Web Server

Steps Followed:

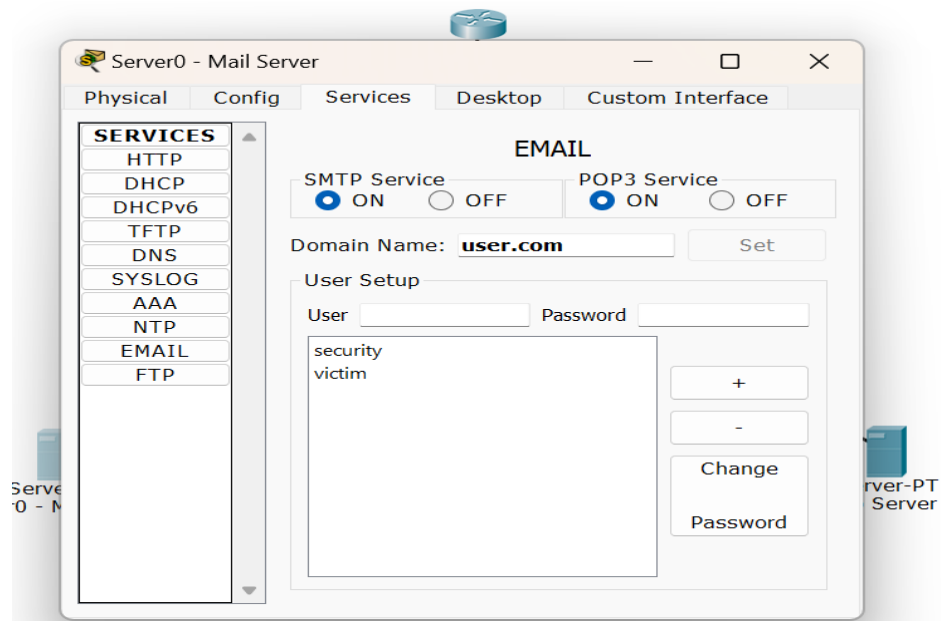
1. **Network Topology:** Connected all devices using a switch.
2. **Mail Server:** Enabled SMTP & POP3, created users.
3. **Web Server:** Created `facebook.html` (fake login page) and `hacked.html`.
4. **PC Configuration:**
 - PC0 configured to receive emails.
 - PC1 sends phishing email to victim.
5. **Victim Interaction:**
 - PC0 opens phishing mail.
 - Clicks on fake link, enters details on fake login page.
 - Form redirects to “You’ve been hacked!” message.

6. Screenshot (Cisco Packet Tracer):

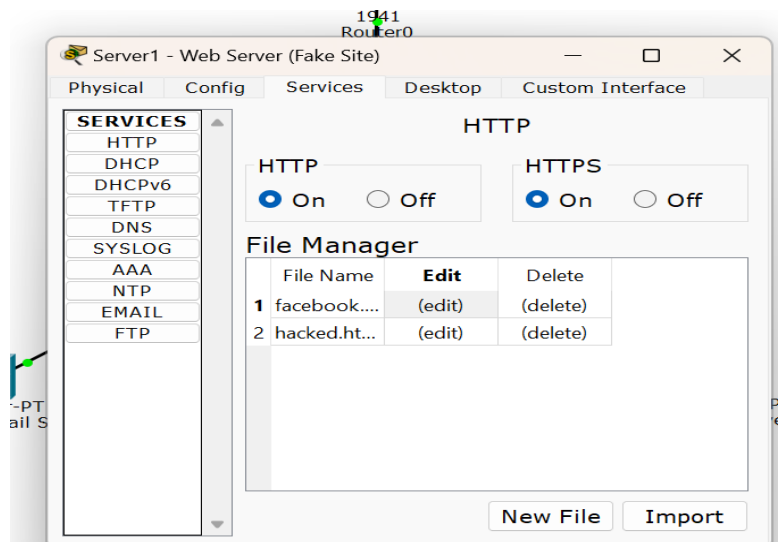
1. Network Topology



2. Email Server Configuration



3. Web Server (facebook.html and hacked.html)



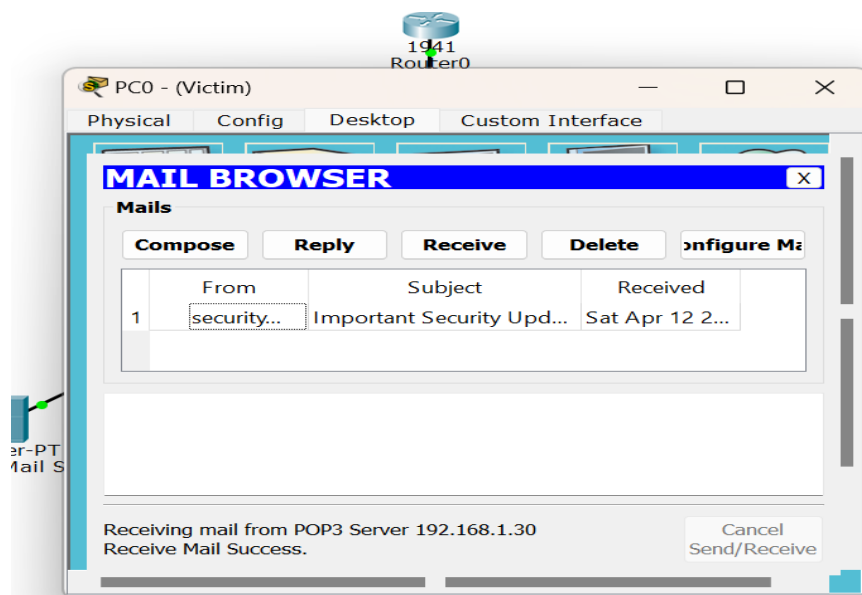
1. File Name :facebook.html

```
<html>
<head>
<title>Facebook ? log in or sign up</title>
<style>
body {
font-family: Arial, sans-serif;
background-color: #e9ebee;
text-align: center;
padding-top: 100px;
}
.login-box {
background: white;
padding: 20px;
width: 300px;
margin: auto;
box-shadow: 0 0 10px rgba(0,0,0,0.3);
}
input[type=text], input[type=password] {
width: 90%;
padding: 10px;
margin: 10px 0;
}
input[type=submit] {
background-color: #1877f2;
color: white;
border: none;
padding: 10px;
width: 95%;
cursor: pointer;
}
</style>
</head>
<body>
<div class="login-box">
<h2>Log in to Facebook</h2>
<form action="hacked.html" method="get">
<input type="text" name="email" placeholder="Email or Phone"><br>
<input type="password" name="pass" placeholder="Password"><br>
<input type="submit" value="Log In">
</form>
</div>
</body>
</html>
```

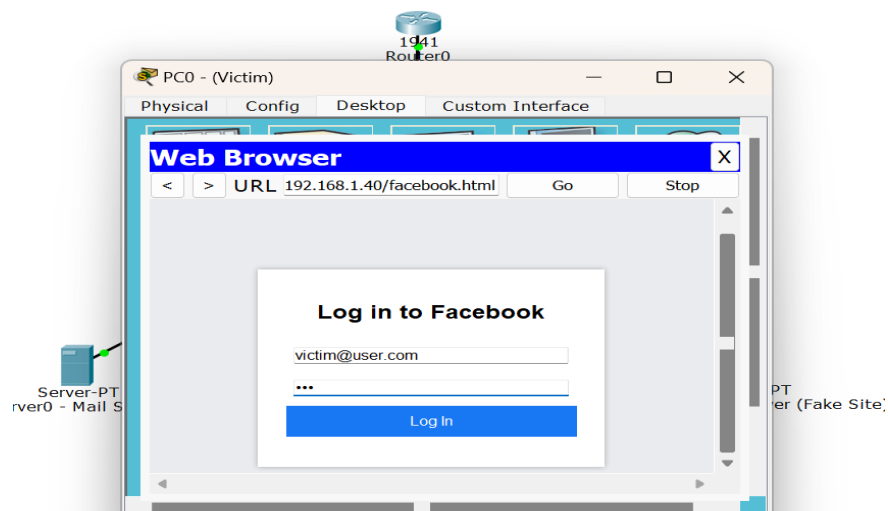

2. File Name : hacked.html

```
<html>
<head>
<title>Hacked!</title>
</head>
<body style="background-color:black; color:red; text-align:center; padding-top:100px;">
<h1>?? You've been hacked!</h1>
<p>Your credentials were sent to the attacker.</p>
</body>
</html>
```

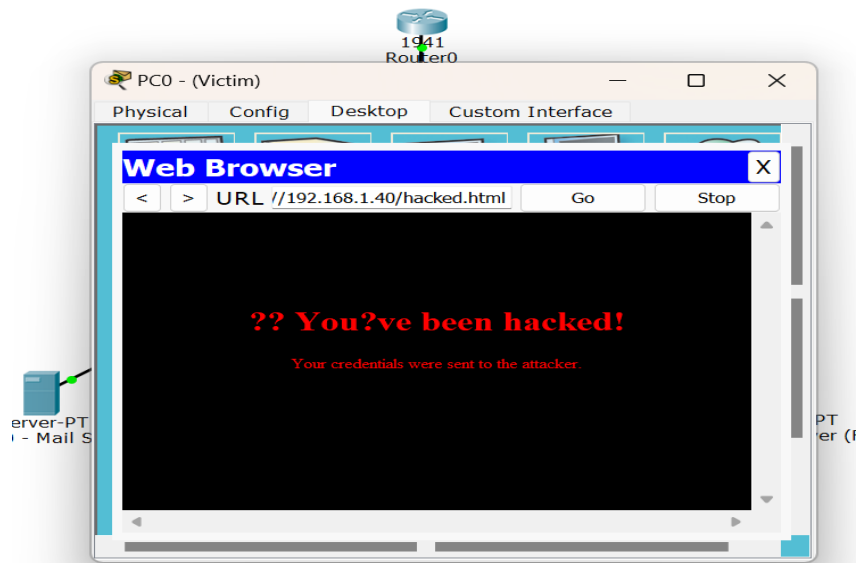
4. PC0 receiving phishing email



5. Victim browsing the fake login page



6. Final “hacked” message on screen



7. Conclusion:

This simulation successfully demonstrated a phishing attack scenario using Cisco Packet Tracer. The project helped in understanding how phishing works and highlighted the importance of cybersecurity awareness. Even though this was a controlled and safe environment, it mirrors real-world scenarios where unsuspecting users can be tricked into giving away their credentials. Through this project, learners gain insight into the techniques used by attackers and how to defend against them.